

EA-0031 - AQELYN Engineering Archive

AQELYN - EA-0031 Engineering Archive

IS-031 - AQELYN Data Security Posture Management (DSPM) Intelligence Engine

****Archive ID:**** EA-0031

****Implementation Specification:**** IS-031

****Component:**** AQELYN Data Security Posture Management (DSPM) Intelligence Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Predecessor Archives:**** EA-0001 through EA-0030

****Next Specification:**** IS-032 - AQELYN Secrets Security & Cryptographic Asset Intelligence Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0031 |

| Specification | IS-031 - AQELYN Data Security Posture Management (DSPM) Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0031.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-031 complete; EA-0031 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

| EA-0004 | IS-004 | AQELYN Evidence Engine |

- | EA-0005 | IS-005 | AQELYN Knowledge Graph |
- | EA-0006 | IS-006 | AQELYN Trust Engine |
- | EA-0007 | IS-007 | AQELYN Mission Engine |
- | EA-0008 | IS-008 | AQELYN Workflow Engine |
- | EA-0009 | IS-009 | AQELYN Policy Engine |
- | EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |
- | EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |
- | EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |
- | EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |
- | EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |
- | EA-0015 | IS-015 | AQELYN Security Operations Engine |
- | EA-0016 | IS-016 | AQELYN Digital Forensics Engine |
- | EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |
- | EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |
- | EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |
- | EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |
- | EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |
- | EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |
- | EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |
- | EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |
- | EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |
- | EA-0026 | IS-026 | AQELYN Configuration Compliance & Drift Intelligence Engine |
- | EA-0027 | IS-027 | AQELYN Identity Threat Detection & Behavioral Analytics Engine |
- | EA-0028 | IS-028 | AQELYN Cloud Security Posture Management Intelligence Engine |
- | EA-0029 | IS-029 | AQELYN SaaS Security Posture Management Intelligence Engine |
- | EA-0030 | IS-030 | AQELYN Software Supply Chain Security & SBOM Intelligence Engine |
- | EA-0031 | IS-031 | AQELYN Data Security Posture Management Intelligence Engine |

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

2. IS-031 Specification Identity

Specification ID: IS-031

Name: AQELYN Data Security Posture Management (DSPM) Intelligence Engine

Engineering Archive Target: EA-0031

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: Complete

Predecessor: IS-030 - AQELYN Software Supply Chain Security & SBOM Intelligence Engine

3. Purpose

The AQELYN Data Security Posture Management (DSPM) Intelligence Engine provides continuous discovery, classification, protection, governance, and risk assessment of enterprise data across cloud, on-premises, SaaS, and hybrid environments.

The engine continuously discovers sensitive data, classifies information assets, evaluates access patterns, detects data exposure risks, validates data protection controls, and correlates data security posture using explainable AI and policy-driven analytics.

It answers:

```
Where is sensitive enterprise data located?  
Which datasets contain regulated information?  
Which data assets are overexposed?  
Which users have excessive access to sensitive data?  
Which data repositories violate security policy?  
Can every data security assessment be explained and audited?
```

4. Mission

The engine shall provide:

```
Continuous data discovery  
Sensitive data classification  
Data exposure analysis  
Data access intelligence  
Data security posture assessment  
Policy compliance validation  
Mission-aware data risk reporting  
Executive data security summaries  
Continuous reassessment  
Enterprise data governance
```

5. Scope

5.1 In Scope

```
Structured databases  
Data warehouses  
Object storage  
File systems  
Cloud storage services  
SaaS data repositories  
Backup repositories  
Data lakes  
Unstructured documents  
Sensitive data classification
```

5.2 Out of Scope

```
Business intelligence reporting  
Application development  
Database performance tuning  
Storage capacity planning  
Data migration projects
```

6. Dependencies

IS-031 depends on:

- IS-001 AQELYN Kernel
- IS-002 Universal Object Model
- IS-003 AQELYN Event Bus
- IS-004 AQELYN Evidence Engine
- IS-005 AQELYN Knowledge Graph
- IS-006 AQELYN Trust Engine
- IS-007 AQELYN Mission Engine
- IS-008 AQELYN Workflow Engine
- IS-009 AQELYN Policy Engine
- IS-010 AQELYN Compliance & Governance Engine
- IS-011 Identity & Access Governance Engine
- IS-012 Asset & Configuration Governance Engine
- IS-013 Risk Intelligence Engine
- IS-014 Threat Intelligence Fusion Engine
- IS-015 Security Operations Engine
- IS-016 Digital Forensics Engine
- IS-017 Threat Detection & Analytics Engine
- IS-018 Automated Response & Orchestration Engine
- IS-019 Security Data Lake & Telemetry Platform
- IS-020 AI Decision Intelligence Engine
- IS-021 Predictive Analytics & Forecasting Engine
- IS-022 Executive Intelligence & Strategic Reporting Engine
- IS-023 Threat Exposure & Attack Surface Management Engine
- IS-024 Vulnerability Intelligence & Prioritization Engine
- IS-025 Cyber Asset Discovery & Inventory Intelligence Engine
- IS-026 Configuration Compliance & Drift Intelligence Engine
- IS-027 Identity Threat Detection & Behavioral Analytics Engine
- IS-028 Cloud Security Posture Management Intelligence Engine
- IS-029 SaaS Security Posture Management Intelligence Engine
- IS-030 Software Supply Chain Security & SBOM Intelligence Engine

7. High-Level Architecture

AQELYN Data Security Posture Management Engine



- Data Discovery Engine
- Sensitive Data Classification Engine
- Data Access Intelligence Engine
- Data Exposure Analysis Engine
- Data Compliance Engine
- Data Risk Engine
- Knowledge Graph Connector
- Security Data Lake Connector
- AI Decision Connector
- Executive Reporting Connector
- Event Publisher

8. Functional Requirements

FR-031-001 - Data Discovery

The engine shall continuously discover:

- Databases
- Storage accounts
- Data lakes
- File shares
- Documents
- Backups
- Archives
- Cloud storage resources

FR-031-002 - Sensitive Data Classification

Continuously identify and classify:

Personally Identifiable Information (PII)
Protected Health Information (PHI)
Payment Card Information (PCI)
Confidential business data
Intellectual property
Source code
Credentials
Custom enterprise classifications

FR-031-003 - Data Risk Detection

Detect:

Publicly exposed storage
Over-permissive access
Sensitive data leakage
Weak encryption
Policy violations
Dormant sensitive repositories
Unauthorized sharing
Compliance gaps

FR-031-004 - Explainable Data Security Intelligence

Every assessment shall include:

Evidence references
Confidence indicators
Policy rationale
Classification evidence
Historical context
Risk explanation

FR-031-005 - Governance

Support:

Data governance
Classification governance
Approval workflows
Policy validation
Auditability
Executive review

FR-031-006 - Event Publication

Publish standardized events:

data.asset.discovered
data.classification.updated
data.exposure.detected
data.policy.violation
data.risk.updated
data.remediation.recommended

9. Non-Functional Requirements

The engine shall provide:

Continuous assessment
Enterprise scalability
Low-latency data analysis
Explainability
Auditability
Repository stability
Backward compatibility

10. Core DSPM Assessment Lifecycle

```

Data Asset Discovered
↓
Sensitive Data Classification
↓
Access Analysis
↓
Exposure Assessment
↓
Risk Assessment
↓
Policy Validation
↓
Continuous Monitoring
  
```

11. Internal Component Architecture

The AQELYN Data Security Posture Management (DSPM) Intelligence Engine is implemented as a modular enterprise data security platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Identity Governance Engine, Cloud Security Posture Management Engine, SaaS Security Posture Management Engine, Risk Intelligence Engine, AI Decision Intelligence Engine, Executive Intelligence Engine, and Security Operations Engine.

```

AQELYN Data Security Posture Management Engine
■
■■■ Data Discovery Engine
■■■ Sensitive Data Classification Engine
■■■ Data Access Intelligence Engine
■■■ Data Exposure Analysis Engine
■■■ Data Compliance Engine
■■■ Data Risk Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher
  
```

12. Component Specifications

12.1 Data Discovery Engine

Continuously discovers enterprise data assets.

Capabilities:

```

Database discovery
File system discovery
Object storage discovery
Data lake discovery
Backup discovery
Repository inventory
  
```

12.2 Sensitive Data Classification Engine

Identifies and classifies sensitive information.

Supports:

```

PII classification
PHI classification
PCI classification
Intellectual property detection
  
```

```
Credential detection  
Custom classification policies
```

12.3 Data Access Intelligence Engine

Analyzes access rights and usage patterns.

Produces:

```
Permission analysis  
Access path analysis  
Privilege evaluation  
Identity correlation  
Access risk scoring
```

12.4 Data Exposure Analysis Engine

Evaluates enterprise data exposure.

Produces:

```
Public exposure detection  
External sharing analysis  
Encryption assessment  
Repository exposure scoring  
Data leakage indicators
```

12.5 Data Compliance Engine

Validates enterprise data against security and regulatory requirements.

Supports:

```
Compliance scoring  
Policy validation  
Control mapping  
Regulatory alignment  
Audit evidence
```

12.6 Data Risk Engine

Calculates enterprise data security risk.

Produces:

```
Data risk score  
Mission impact  
Business impact  
Threat likelihood  
Remediation priority
```

13. Universal Object Model Extensions

13.1 DataAsset

```
DataAsset:  
asset_id  
asset_type  
location  
owner
```

13.2 DataClassification

```
DataClassification:  
classification_id  
sensitivity  
confidence  
classified_at
```

13.3 DataExposure

```
DataExposure:  
exposure_id  
exposure_type  
severity  
detected_at
```

13.4 DataRemediation

```
DataRemediation:  
remediation_id  
recommendation  
owner  
target_date
```

14. Knowledge Graph Integration

Relationships:

```
DataAsset  
↓  
classified_as  
↓  
DataClassification  
  
DataAsset  
↓  
accessed_by  
↓  
Identity  
  
Identity  
↓  
creates  
↓  
DataExposure  
  
DataExposure  
↓  
requires  
↓  
DataRemediation
```

15. Event Bus Integration

15.1 Discovery Events

```
data.asset.discovered  
data.asset.updated  
data.asset.removed
```

15.2 Classification Events

```
data.classification.updated  
data.classification.validated
```


15.3 Exposure Events

```
data.exposure.detected  
data.exposure.resolved
```

15.4 Governance Events

```
data.policy.violation  
data.risk.updated  
data.remediation.recommended
```

16. Security Data Lake Integration

Consumes:

```
Database metadata  
Storage metadata  
Access logs  
Audit logs  
Classification metadata  
Repository telemetry
```

17. AI Decision Intelligence Integration

Consumes:

```
Data risk scores  
Classification intelligence  
Compliance trends  
Remediation recommendations  
Confidence indicators
```

18. Risk & Threat Integration

Consumes:

```
Threat intelligence  
Mission impact  
Business criticality  
Identity risk  
Cloud posture intelligence  
SaaS posture intelligence
```

19. Compliance Integration

Supports:

```
Data governance  
Classification governance  
Policy validation  
Audit evidence  
Executive reporting
```

20. Public APIs

20.1 Data Inventory API

```
GET /data/assets
GET /data/assets/{id}
```

20.2 Classification API

```
GET /data/classifications
GET /data/classifications/{id}
```

20.3 Exposure API

```
GET /data/exposures
GET /data/exposures/{id}
```

20.4 Remediation API

```
GET /data/remediations
POST /data/remediations
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
  ■■■ src/
  ■ ■■■ data_security_posture/
  ■■■ tests/
  ■ ■■■ data_security_posture/
  ■■■ docs/
  ■ ■■■ data_security_posture/
  ■■■ api/
  ■ ■■■ data_security_posture/
  ■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Data Security Posture Management (DSPM) Intelligence Engine is the trusted subsystem responsible for continuously discovering enterprise data assets, classifying sensitive information, analyzing data access, detecting exposure risks, validating data protection controls, and governing enterprise data security posture.

Every data security assessment shall be:

```
Explainable
Evidence-backed
Policy-governed
Data-aware
Risk-aware
Mission-aware
Fully auditable
Continuously reassessed
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Continuous Data Security Assessment
Secure by Design
Policy Enforcement
Explainable Data Intelligence
```

Continuous Monitoring

22.2 Authorization Model

Supported operational roles:

Data Security Administrator
Data Owner
Database Administrator
Security Administrator
SOC Analyst
Compliance Officer
Mission Owner
Executive Reviewer

All data security posture assessments and remediation decisions shall be governed through the AQELYN Policy Engine.

22.3 Data Assessment Integrity

Data security assessments shall maintain:

Unique assessment identifier
Data asset reference
Classification reference
Exposure reference
Evidence references
Risk score
Confidence indicators
Remediation guidance
Audit trail

Assessment history shall be append-only.

22.4 Classification Integrity

Data classification records shall maintain:

Classification identifier
Sensitivity category
Detection method
Confidence score
Classification timestamp
Evidence references
Policy references
Audit record

No classification shall be considered authoritative without evidence references, confidence indicators, and validation state.

23. Data Security Lifecycle

23.1 Assessment Lifecycle

Data Asset Discovered
↓
Sensitive Data Classification
↓
Access Analysis
↓
Exposure Assessment
↓
Risk Assessment
↓
Continuous Monitoring

23.2 Remediation Lifecycle

```
graph TD; A[Issue Detected] --> B[Risk Prioritized]; B --> C[Remediation Generated]; C --> D[Approval Granted]; D --> E[Remediation Applied]; E --> F[Validation Completed];
```

23.3 Audit Lifecycle

```
graph TD; A[Assessment Created] --> B[Evidence Linked]; B --> C[Policy Validated]; C --> D[Audit Stored];
```

24. Continuous DSPM Operations

The engine continuously evaluates:

```
graph TD; A[Enterprise databases] --> B[Cloud storage]; B --> C[Object storage]; C --> D[File systems]; D --> E[Data lakes]; E --> F[Backup repositories]; F --> G[Sensitive datasets]; G --> H[Access permissions];
```

25. Performance Requirements

The engine shall support:

```
graph TD; A[Continuous assessment] --> B[Low-latency data analysis]; B --> C[Enterprise-scale repositories]; C --> D[Concurrent data classification]; D --> E[High availability]; E --> F[Continuous operation];
```

26. Scalability Requirements

The engine shall scale to support:

```
graph TD; A[Global enterprises] --> B[Billions of records]; B --> C[Millions of files]; C --> D[Thousands of repositories]; D --> E[Distributed data assessments]; E --> F[Long-term posture history];
```

27. Audit Requirements

Every DSPM operation shall generate immutable audit records.

Audit events include:

- Data asset discovered
- Classification updated
- Exposure detected
- Policy violation identified
- Risk updated
- Remediation completed

28. Failure Handling

28.1 Discovery Failure

- Discovery retried
- Failure recorded
- Administrator notified

28.2 Classification Failure

- Classification retried
- Previous assessment retained
- Audit generated

28.3 Exposure Analysis Failure

- Analysis repeated
- Manual review initiated
- Audit recorded

28.4 Policy Failure

- Data access blocked
- Policy violation recorded
- Manual approval required

29. Testing Strategy

29.1 Unit Testing

Validate:

- Data Discovery Engine
- Sensitive Data Classification Engine
- Data Access Intelligence Engine
- Data Exposure Analysis Engine
- Data Compliance Engine
- Data Risk Engine

29.2 Integration Testing

Verify interaction with:

- Kernel
- Knowledge Graph
- Security Data Lake
- Identity Governance Engine
- Cloud Security Posture Management Engine
- SaaS Security Posture Management Engine
- Risk Intelligence Engine
- AI Decision Engine

Executive Reporting Engine
Security Operations Engine

29.3 System Testing

Validate:

Data discovery
Classification
Exposure analysis
Access intelligence
Risk scoring
Audit generation

29.4 Security Testing

Verify:

Authorization
Policy enforcement
Explainability
Audit logging
Data security assessment integrity

29.5 Regression Testing

Verify IS-001 through IS-030 remain unaffected.

30. Acceptance Criteria

IS-031 is complete when:

Data Discovery Engine implemented
Sensitive Data Classification Engine implemented
Data Access Intelligence Engine implemented
Data Exposure Analysis Engine implemented
Data Compliance Engine implemented
Data Risk Engine implemented
Repository unchanged
Testing documented

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/  
■■■ src/data_security_posture/  
■■■ tests/data_security_posture/  
■■■ docs/data_security_posture/  
■■■ api/data_security_posture/  
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-031 introduces the AQELYN Data Security Posture Management (DSPM) Intelligence Engine, providing enterprise-scale data discovery, sensitive data classification, access intelligence, exposure analysis, explainable data risk scoring, compliance validation, and continuous enterprise data governance.

Major capabilities include:

- Data Discovery
- Sensitive Data Classification
- Data Access Intelligence
- Data Exposure Analysis
- Data Compliance Validation
- Data Risk Scoring
- Mission-Aware Data Risk Reporting
- Executive Data Security Reporting
- Continuous Data Monitoring
- Enterprise Data Governance

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

Specification ID : IS-031
Title : AQELYN Data Security Posture Management (DSPM) Intelligence Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0031

34. EA-0031 Engineering Objective

The objective of IS-031 was to introduce a dedicated Data Security Posture Management Intelligence Engine that enables AQELYN to continuously discover enterprise data assets, classify sensitive data, analyze access paths, detect data exposure, validate data compliance, score data risk, and generate remediation recommendations.

The engine extends AQELYN from software supply chain governance into enterprise data security posture and data governance.

35. EA-0031 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Data discovery
- Sensitive data classification
- Data access intelligence
- Data exposure analysis
- Data compliance validation
- Data risk scoring
- Data remediation recommendations
- Knowledge Graph integration
- Security Data Lake integration
- Risk and threat intelligence integration
- AI Decision integration
- Executive reporting integration
- Event publishing

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated DSPM Engine

Data security posture management responsibilities are implemented as a standalone engine rather than embedded in cloud posture, SaaS posture, or compliance governance.

Rationale:

- Clear separation of data security posture from infrastructure posture.
- Independent lifecycle and governance.
- Better support for sensitive data classification, access analysis, and exposure detection.
- Improved traceability for data posture assessments.

36.2 Decision 2 - Data Classification Is Evidence-Backed

Data classification records are modeled with sensitivity, confidence, evidence, detection method, and validation state.

Benefits:

- Classification decisions become auditable.
- False positives and false negatives can be reviewed.
- Regulatory reporting can cite classification evidence.

36.3 Decision 3 - Data Access Is Risk-Correlated

Access analysis correlates identities, permissions, access paths, and data sensitivity.

Benefits:

- Over-permissioned access becomes visible.
- Identity risk can be connected to sensitive data exposure.
- Data remediation can be prioritized based on business and mission impact.

36.4 Decision 4 - Data Exposure Is Continuously Assessed

Public access, external sharing, weak encryption, dormant sensitive repositories, and compliance gaps are continuously evaluated.

Benefits:

- Sensitive data exposure can be detected before exploitation.
- Data security posture becomes measurable.
- Executives can track data exposure trends.

36.5 Decision 5 - Event-Driven Data Security Lifecycle

Data discovery, classification, exposure, risk, policy, and remediation events are published through the AQELYN Event Bus.

Examples include:

- data.asset.discovered
- data.asset.updated
- data.asset.removed
- data.classification.updated
- data.classification.validated
- data.exposure.detected
- data.exposure.resolved
- data.policy.violation
- data.risk.updated
- data.remediation.recommended

36.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```

DataAsset
DataClassification
DataExposure
DataRemediation

```

37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Data asset, classification, exposure, remediation objects |

| IS-003 Event Bus | Data discovery, classification, exposure, policy, risk events |

| IS-004 Evidence Engine | Evidence references and data security support |

| IS-005 Knowledge Graph | Data, classification, identity, exposure, remediation relationships |

| IS-006 Trust Engine | Data confidence and classification trust |

| IS-007 Mission Engine | Mission-aware data risk prioritization |

| IS-008 Workflow Engine | Data remediation and approval workflows |

| IS-009 Policy Engine | Data governance and policy validation |

| IS-010 Compliance Engine | Data governance, classification governance, audit reporting |

| IS-011 Identity Governance Engine | Identity permissions and data access context |

| IS-013 Risk Intelligence Engine | Risk scoring and business impact |

| IS-014 Threat Intelligence Engine | Threat context for data exposure |

| IS-015 Security Operations Engine | Data exposure investigations and SOC workflows |

| IS-019 Security Data Lake | Database metadata, storage metadata, access logs, classification metadata |

| IS-020 AI Decision Engine | Remediation recommendations and confidence scoring |

| IS-022 Executive Reporting Engine | Executive data security summaries |

| IS-028 Cloud Security Posture Engine | Cloud storage and cloud posture context |

| IS-029 SaaS Security Posture Engine | SaaS data repository and tenant context |

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```

AQELYN/
■■■ src/data_security_posture/
■■■ tests/data_security_posture/
■■■ api/data_security_posture/
■■■ docs/data_security_posture/
■■■ archive/

```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces data-security-posture-specific security controls:

- Policy-governed data assessments
- Evidence-backed classification
- Continuous data exposure detection
- Data access intelligence
- Sensitive repository risk scoring
- Compliance validation
- Data remediation ownership tracking
- Data assessment audit trail
- Role-authorized DSPM administration

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

- Continuous data discovery
- Sensitive data classification
- Data exposure analysis
- Data access intelligence
- Data security posture assessment
- Policy compliance validation
- Mission-aware data risk reporting
- Executive data security summaries
- Continuous reassessment
- Enterprise data governance

41. Risks Identified

| Risk | Mitigation |

|---|---|

| Incomplete data discovery | Multiple repository discovery mechanisms and continuous assessment |

| False data classification | Confidence indicators and classification validation |

| Over-permissive data access | Data Access Intelligence Engine |

| Public sensitive data exposure | Data Exposure Analysis Engine |

| Weak encryption | Data protection validation and remediation |

| Dormant sensitive repositories | Continuous repository evaluation |

| Poor auditability | Evidence-backed assessments and immutable audit events |

| Unauthorized remediation | Policy enforcement and workflow approval |

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

- Unit testing
- Integration testing
- System testing
- Security testing
- Regression testing

Acceptance criteria cover data discovery, sensitive data classification, access intelligence, exposure analysis, compliance, data risk, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

- Modular architecture
- Event-driven communication
- Immutable evidence references
- Traceability
- Explainability
- Security by design
- Repository stability
- Backward compatibility
- Governance-before-archive discipline

44. Dependencies

Required:

- EA-0001 through EA-0030
- IS-001 through IS-031

Enables:

- IS-032 and subsequent secrets security, cryptographic asset intelligence, privacy intelligence, and cyber resilience components

45. Completion Record

Engineering Archive : EA-0031
 Implementation Specification : IS-031
 Title : AQELYN Data Security Posture Management (DSPM) Intelligence Engine
 Engineering Status : COMPLETE
 Repository Status : UNCHANGED
 Architecture Status : EXTENDED
 Backward Compatibility : MAINTAINED

46. Archive Index Update

- EA-0001 IS-001 AQELYN Kernel
- EA-0002 IS-002 Universal Object Model
- EA-0003 IS-003 AQELYN Event Bus
- EA-0004 IS-004 AQELYN Evidence Engine
- EA-0005 IS-005 AQELYN Knowledge Graph
- EA-0006 IS-006 AQELYN Trust Engine
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- EA-0024 IS-024 AQELYN Vulnerability Intelligence & Prioritization Engine
- EA-0025 IS-025 AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

```
EA-0026 IS-026 AQELYN Configuration Compliance & Drift Intelligence Engine
EA-0027 IS-027 AQELYN Identity Threat Detection & Behavioral Analytics Engine
EA-0028 IS-028 AQELYN Cloud Security Posture Management Intelligence Engine
EA-0029 IS-029 AQELYN SaaS Security Posture Management Intelligence Engine
EA-0030 IS-030 AQELYN Software Supply Chain Security & SBOM Intelligence Engine
EA-0031 IS-031 AQELYN Data Security Posture Management Intelligence Engine
```

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0031

Current Status:
EA-0031 COMPLETE

Next Implementation Specification:
IS-032 - AQELYN Secrets Security & Cryptographic Asset Intelligence Engine

48. Engineering Archive Publication Standard

EA-0031 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
├──
├── diagrams/
│   ├── Architecture.svg
│   ├── Component.svg
│   ├── Workflow.svg
│   ├── EventFlow.svg
│   └── Integration.svg
├──
├── examples/
│   └── example_*.md
├──
├── HTML/
│   ├── index.html
│   ├── styles.css
│   └── assets/
├──
├── images/
├──
├── journal/
│   └── Engineering_Journal.md
├──
├── manifest/
│   └── manifest.json
├──
├── MD/
│   └── EA-xxxx.md
├──
├── PDF/
│   └── EA-xxxx.pdf
├──
├── requirements/
│   └── Requirements_Matrix.md
├──
├── traceability/
│   └── Traceability_Matrix.md
├──
└── README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-031-001	Continuously discover data assets	Sections 8, 12, 23	Complete
FR-031-002	Classify sensitive data	Sections 8, 12, 23	Complete
FR-031-003	Detect data risks	Sections 8, 12, 23	Complete
FR-031-004	Provide explainable data security intelligence	Sections 8, 22, 36	Complete
FR-031-005	Support data governance	Sections 8, 22, 23	Complete
FR-031-006	Publish data events	Sections 8, 15, 36	Complete
NFR-031-001	Continuous assessment	Sections 9, 24	Complete
NFR-031-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-031-003	Low-latency data analysis	Sections 9, 25	Complete
NFR-031-004	Explainability	Sections 9, 22, 36	Complete
NFR-031-005	Auditability	Sections 9, 27, 39	Complete
NFR-031-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source	Target	Relationship
IS-031 Purpose	EA-0031 Objective	Defines why the engine exists
Data Discovery Engine	FR-031-001	Discovers data assets
Sensitive Data Classification Engine	FR-031-002	Classifies sensitive data
Data Access Intelligence Engine	FR-031-003	Analyzes access risk
Data Exposure Analysis Engine	FR-031-003	Detects exposure risks
Data Compliance Engine		Compliance validation Maps data posture to controls
Data Risk Engine		Risk scoring Calculates data security risk
Event Publisher	FR-031-006	Publishes data events
Security Data Lake Integration		Data telemetry Supplies metadata, logs, classification data
AI Decision Integration		Remediation recommendations Supplies confidence and recommendations
Risk & Threat Integration		Threat context Supplies threat, identity, cloud, SaaS context
Compliance Integration		Governance and audit Supports data governance and evidence
Repository Validation	Repository Standard	Confirms no top-level redesign

51. Engineering Journal

Journal Entry - EA-0031

EA-0031 was created to archive completion of IS-031 - AQELYN Data Security Posture Management Intelligence Engine.

The archive records the expansion of AQELYN into enterprise data security posture management. IS-031 defines the structure needed to discover data assets, classify sensitive data, analyze access, detect exposure, validate compliance, score data risk, generate remediation recommendations, and publish data security events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

DSPM must be modeled separately from cloud posture, SaaS posture, and compliance governance. Cloud and SaaS posture assess platforms and applications; DSPM focuses on sensitive data location, classification, access paths, exposure, and data governance.

Governance Note

EA-0031 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Data Asset

```
asset_id: DATA-0001
asset_type: object_storage_bucket
location: cloud://storage/finance-sensitive
owner: finance_data_owner
```

52.2 Example Data Classification

```
classification_id: CLASS-2001
sensitivity: regulated_pii
confidence: 0.94
classified_at: 2026-07-07T12:00:00Z
```

52.3 Example Data Exposure

```
exposure_id: DEXP-3001
exposure_type: public_storage_access
severity: high
detected_at: 2026-07-07T12:05:00Z
```

52.4 Example DSPM Event

```
{
  "event_type": "data.exposure.detected",
  "asset_id": "DATA-0001",
  "risk_score": 89,
  "source_engine": "aqelyn_data_security_posture_management_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0031.md
PDF/EA-0031.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_data_security_posture.md
```

54. Final Archive Statement

EA-0031 is the Engineering Archive for IS-031 - AQELYN Data Security Posture Management (DSPM) Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

EA-0031 COMPLETE
IS-031 COMPLETE
NEXT: IS-032